Technische Hochschule Ingolstadt
Fakultät Maschinenbau

Autonomes Fliegen

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Inhalt der Vorlesung 1/2

Kapitel

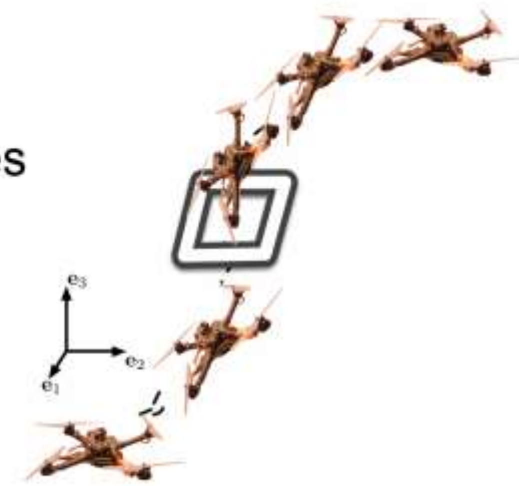
- **Kapitel 1: Übersicht über Unmanned Aerial Systems (UAVs) (1 W)**
- **Kapitel 2: Wesentliche Komponenten/Subsysteme eines UAV (1 W)**
- **Kapitel 3: Modellierung und Simulation von UAVs (1 W)**
- **Kapitel 4: Flight State Controller (PD Cascade) (1W)**
- **Kapitel 5: Pfadplanung (4 W)**
- **Kapitel 6: Perception und Sensing (1 W)**
- **Kapitel 7: Navigation (2 W)**
- **Kapitel 8: Multi-Vehicle Cooperation and Coordination (1 W)**

References

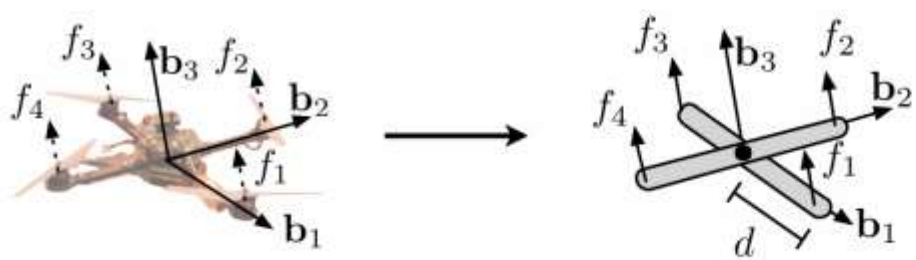
- D. Mellinger, "Trajectory Generation and Control for Quadrotors" Ph.D. thesis, University of Pennsylvania, Philadelphia, USA, 2012.
- Modelling and Control of Mini-Flying Machines, 2005.
- D. Mellinger and V. Kumar, "Minimum snap trajectory generation and control for quadrotors," 2011 IEEE International Conference on Robotics and Automation, Shanghai, China, 2011, pp. 2520-2525, doi: 10.1109/ICRA.2011.5980409.
- ❑ D. McLean, Automatic Flight Control Systems, Prentice Hall International (UK) Ltd, Cambridge, Great Britain, 1990.

Outline

- Modeling:
  - Dynamic model from first principles
  - Propeller model and force and moments generation
- Control
  - Attitude control (inner loop)
  - Position control (outer loop)
- Current research challenges



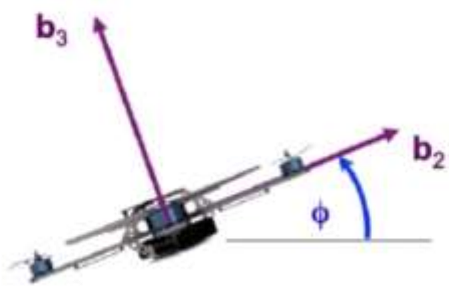
1. Vehicle model
2. Attitude and position control
3. Trajectory generation



D. Mellinger, N. Michael, and V. Kumar. Trajectory generation and control for precise aggressive maneuvers with quadrotors. Intl. J. Robot. Research, 31(5):664–674, Apr. 2012

Planner Quadrotor Model

- **2D Equations of Motion**



$$\ddot{y} = -\frac{u_1}{m}\sin(\phi)$$
$$\ddot{z} = -g + \frac{u_1}{m}\cos(\phi)$$
$$\ddot{\phi} = \frac{u_2}{I_{xx}}$$

$$\begin{bmatrix} \ddot{y} \\ \ddot{z} \\ \ddot{\phi} \end{bmatrix} = \begin{bmatrix} 0 \\ -g \\ 0 \end{bmatrix} + \begin{bmatrix} -\frac{1}{m}\sin\phi & 0 \\ \frac{1}{m}\cos\phi & 0 \\ 0 & \frac{1}{I_{xx}} \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$$

Planner Quadrotor Model

- **Equilibrium Hover Configuration**

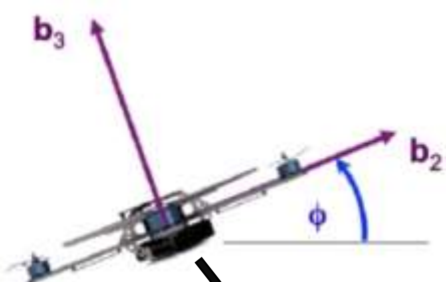
$$y_0, z_0, \phi_0, u_1 = mg, u_2 = 0$$

- **Linearized Dynamics**

$$\begin{aligned} \ddot{y} &= -g\phi \\ \ddot{z} &= -g + \frac{u_1}{m} \\ \ddot{\phi} &= \frac{u_2}{I_{xx}} \end{aligned}$$

Tracking

- **Given** $r_d = \begin{bmatrix} y(t) \\ z(t) \end{bmatrix}$



$r_d - r$

$r_d(t)$

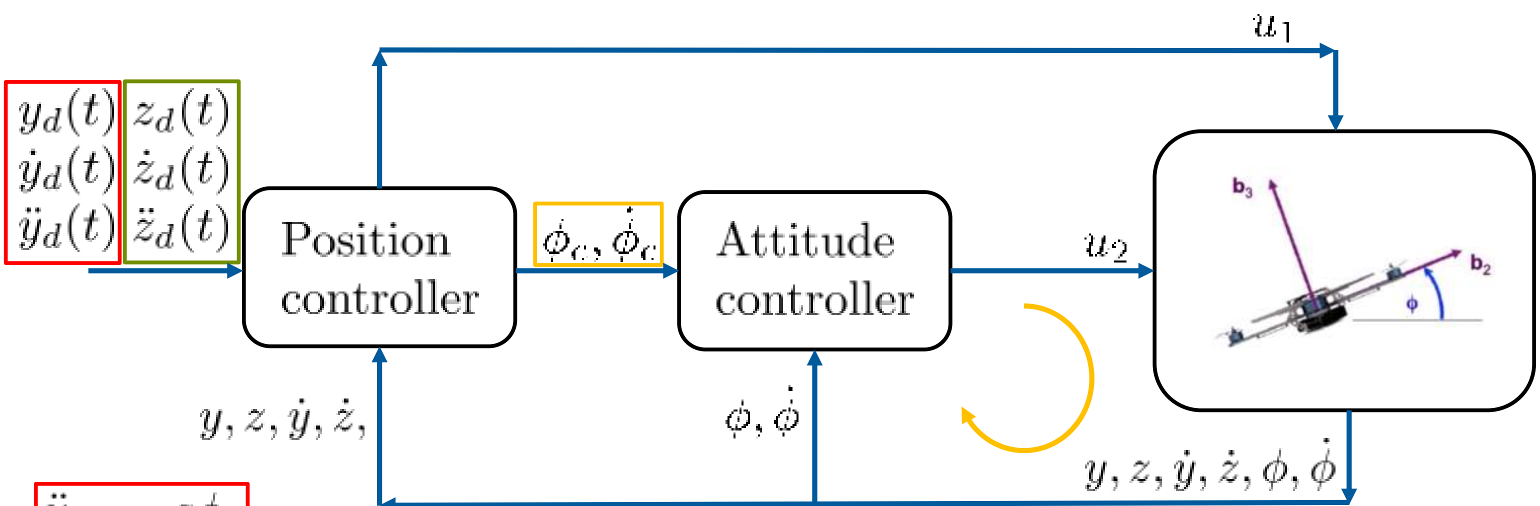
desired trajectory
(position, velocity, acceleration)

Want

$$
\begin{aligned}
e_p &= r_d(t) - r \\
e_v &= \dot{r}_d(t) - \dot{r} \\
(\ddot{r}_d(t) - \ddot{r}_c) &+ k_d e_v + k_p e_p = 0
\end{aligned}
$$

Commanded acceleration, calculated by the controller

2D Controller



$$\boxed{\ddot{y} = -g\phi}$$

$$\boxed{\ddot{z} = -g + \frac{u_1}{m}}$$

$$\boxed{\ddot{\phi} = \frac{u_2}{I_{xx}}}$$

$$\boxed{u_1 = m(g + \ddot{z}_d + k_{d,z}(\dot{z}_d - \dot{z}) + k_{p,z}(z_d - z))}$$

$$\boxed{u_2 = I_{xx}(\ddot{\phi}_c + k_{d,\phi}(\dot{\phi}_c - \dot{\phi}) + k_{p,z}(\phi_c - \phi))}$$

$$\boxed{\phi_c = -\frac{1}{g}(\ddot{y}_d + k_{d,y}(\dot{y}_d - \dot{y}) + k_{p,y}(y_d - y))}$$

Hovering

This control law works well under hover conditions. Linearizing the dynamic model at the hover configuration, where we have: $u_1 \approx mg$, $\theta \approx 0$, $\phi \approx 0$, $\psi \approx \psi_0$, $u_2 \approx 0$, $u_2 \approx 0$, $p \approx 0$, $q \approx 0$ and $r \approx 0$. Under these approximations, the system model is reduced to:

$$m \begin{bmatrix} \ddot{x} \\ \ddot{y} \\ \ddot{z} \end{bmatrix} = \begin{bmatrix} (c\psi s\theta + c\theta s\phi s\psi)u_1 \\ (s\psi s\theta - c\theta s\phi c\psi)u_1 \\ -mg + c\phi c\theta u_1 \end{bmatrix}$$

$$I \begin{bmatrix} \dot{p} \\ \dot{q} \\ \dot{r} \end{bmatrix} = u_2$$

Control

Proportional Derivative (PD) Control

The reference trajectory is: $r_{ref} = \left[\begin{array}{cccc} x_{des} & y_{des} & z_{des} & \psi_{des} \end{array}\right]^T$ The commanded linear accelerations can be calculated as:

$$\ddot{x}_c = \ddot{x}_{des} + k_{dx}(\dot{x}_{des} - \dot{x}) + k_{px}(x_{des} - x)$$
$$\ddot{y}_c = \ddot{y}_{des} + k_{dy}(\dot{y}_{des} - \dot{y}) + k_{py}(y_{des} - y)$$
$$\ddot{z}_c = \ddot{z}_{des} + k_{dz}(\dot{z}_{des} - \dot{z}) + k_{pz}(z_{des} - z)$$

The commanded roll, pitch and yaw are:

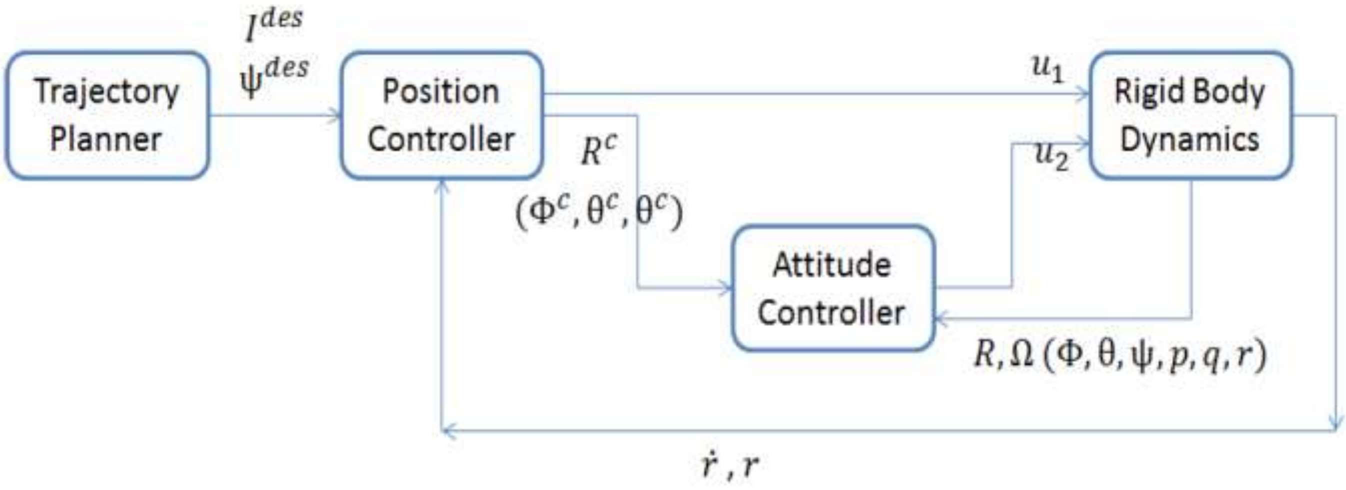
$$\phi_c = \frac{1}{g}(\ddot{x}_c sin(\psi_{des}) - \ddot{y}_c cos(\psi_{des}))$$
$$\theta_c = \frac{1}{g}(\ddot{x}_c cos(\psi_{des}) + \ddot{y}_c sin(\psi_{des}))$$
$$\psi_c = \psi_{des}$$

Control Commands

$$u_1 = m(g + \ddot{z}_c)$$

$$u_2 = \mathbf{I} \begin{bmatrix} k_{p\phi}(\phi_c - \phi) + k_{d\phi}(p_c - p) \\ k_{p\theta}(\theta_c - \theta) + k_{d\theta}(q_c - q) \\ k_{p\psi}(\psi_c - \psi) + k_{d\theta}(r_c - r) \end{bmatrix}$$

Control Structure



Control Block Diagram